

Musculoskeletal System and Exercise

Dr. Praveen Kumar T K

Associate Professor
Department of Physical Education
College of Engineering Trivandrum



Overview of the Musculoskeletal System

- The musculoskeletal system provides structure and support to the body.
- Consists of bones, muscles, cartilage, tendons, ligaments, and other connective tissues.
- Enables movement, protection of organs, and production of red blood cells.



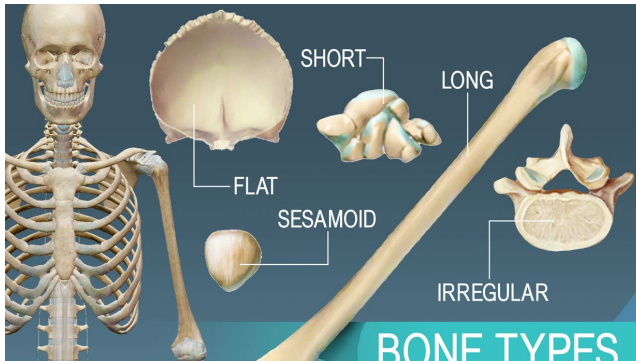
Functions of the Musculoskeletal System

- Support: Provides a framework to support the body.
- Movement: Enables movement through the contraction of muscles.
- Protection: Protects vital organs like the heart, lungs, and brain.
- Mineral storage: Stores essential minerals such as calcium and phosphorus.
- Blood cell production: Bone marrow produces red and white blood cells.



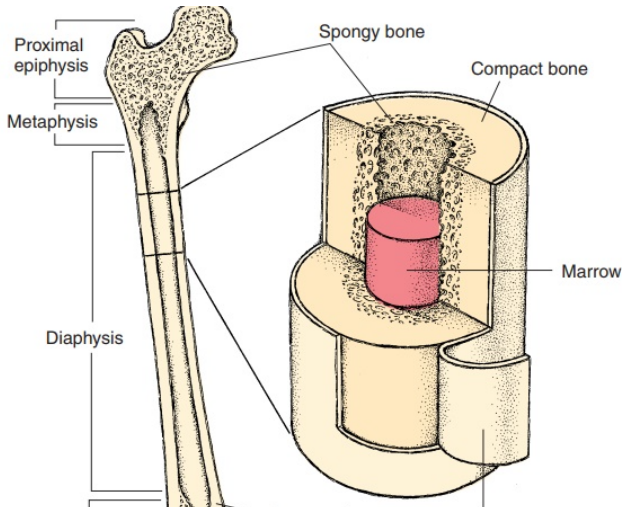
Types of Bones

- Long bones (e.g., femur)
- Short bones (e.g., carpals)
- Flat bones (e.g., skull)
- Irregular bones (e.g., vertebrae)
- Sesamoid bones (e.g., patella)



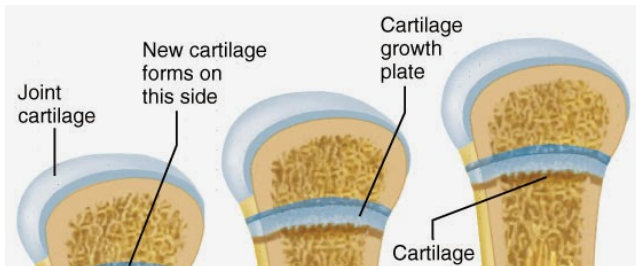
Structure of Bones

- Compact bone: Dense outer layer.
- Spongy bone: Lighter, porous inner structure.



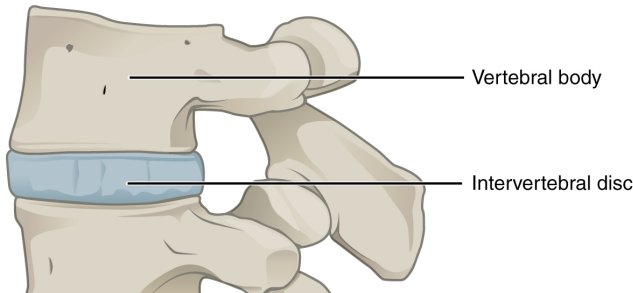
Articular System

- The articular system, also known as the joint system, is a crucial component of the human skeletal system that comprises all the joints in the body.
- **Articular Cartilage:**
 - A smooth, white tissue that covers the ends of bones where they come together to form joints.
- **Growth Cartilage:**
 - Also known as epiphyseal cartilage or growth plate, found at the ends of long bones in children and adolescents.



Fibrocartilage

- Fibrocartilage is a tough, dense connective tissue found in areas requiring support and the ability to withstand pressure.
- Common locations include:
 - Intervertebral discs (spinal column)
 - Pubic symphysis (pelvic region)
 - Menisci of the knee joint
 - Labrum of the shoulder and hip joints

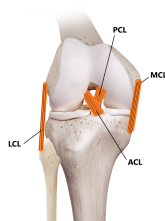


Tendons and Ligaments

- Tendons: Connect muscles to bones.



- Ligaments: Connect bones to other bones.
- Both are made of tough, fibrous tissue.
- Provide stability to joints.



Joints of the Body

- Joints allow for movement between bones.
- Types of joints:
 - Synovial joints (e.g., knee, shoulder)
 - Cartilaginous joints (e.g., between vertebrae)
 - Fibrous joints (e.g., skull sutures)



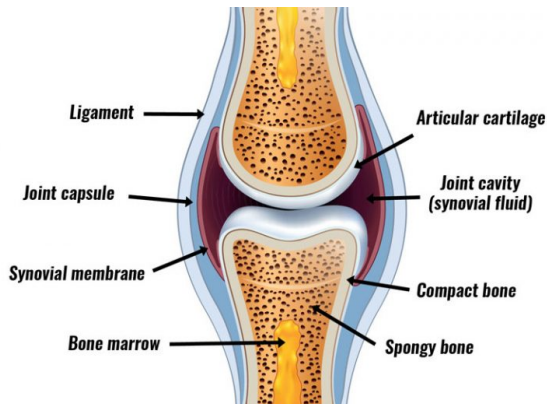
Synovial Joints

- Most common type of joint.
- Contains synovial fluid for lubrication.
- Allows for a wide range of movements:
 - Hinge joint (e.g., elbow)
 - Ball-and-socket joint (e.g., shoulder, hip)
 - Pivot joint (e.g., neck)



Synovial Joints

- Most common type of joint.



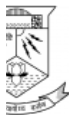
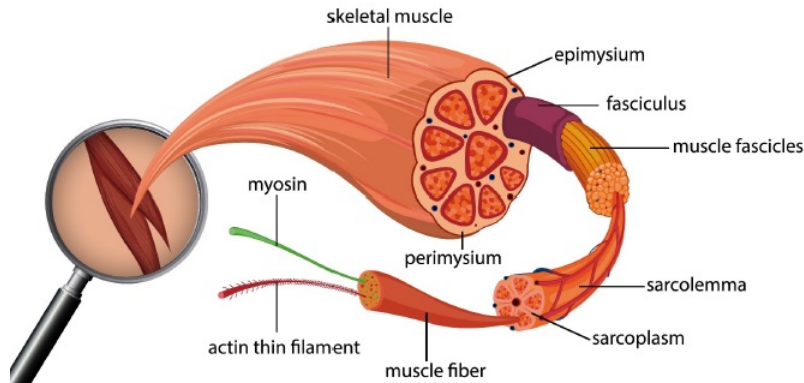
Types of Muscles

- Skeletal muscles: Voluntary muscles that attach to bones.
- Cardiac muscles: Involuntary muscles found in the heart.
- Smooth muscles: Involuntary muscles found in organs such as the stomach and blood vessels.



Skeletal Muscle Structure

- Muscle fibers: Long, cylindrical cells. A single muscle cell is known as a muscle fiber.
- It has a cell membrane and the same organelles as other cell types but is uniquely multinucleated.



Types of Muscle Fibers - Overview

- A single skeletal muscle contains fibers with different speeds of shortening and force generation.
- Muscle fibers are classified into two major types:
 - Type I (slow or slow-twitch) fibers
 - Type II (fast or fast-twitch) fibers
- Type I fibers take approximately 110 ms to reach peak tension.
- Type II fibers can reach peak tension in about 50 ms.



Type II Muscle Fibers

- Type II fibers are divided into:
 - Fast-twitch Type IIa fibers
 - Fast-twitch Type IIx fibers
- Type IIa fibers are the most frequently recruited during high-intensity exercise.
- Most muscles are composed of:
 - 50% Type I fibers
 - 25% Type IIa fibers
 - 25% Type IIx fibers (with 1-3% Type IIc fibers)



Characteristics of Muscle Fibers

- Type I fibers have higher aerobic endurance and are suited for low-intensity endurance activities.
- Type II fibers are suited for anaerobic activity:
 - Type IIa fibers play a major role in high-intensity exercise.
 - Type IIx fibers are recruited when high force is demanded.
- Type II fibers are larger than Type I fibers and contain more muscle fibers per motor unit.



Overview of Muscle Contraction

- Muscle movement can be categorized into three types of contractions:
 - Concentric
 - Isometric/static
 - Eccentric
- In activities like running and jumping, all three contractions may occur in a smooth, coordinated movement.



Concentric and Isometric Contractions

- **Concentric Contraction:**

- The principal action of a muscle, shortening, is called a concentric contraction.
- Thin filaments are pulled toward the center of the sarcomere.
- Joint movement is produced; hence, it's considered a dynamic contraction.

- **Isometric/Static Contraction:**

- Muscles can act without moving; they generate force while remaining static (unchanged).
- Joint angle does not change during static contraction.



Eccentric Contraction

- Muscles can exert force while lengthening, known as eccentric contraction.
- This movement is also a dynamic contraction since joint movement occurs.
- Example: The biceps brachii extending the elbow to lower a heavy weight.
- In this case, thin filaments are pulled farther from the center of the sarcomere, stretching it.



Muscular Adaptations Due to Aerobic Training

- **Increase in Type 1 Muscle Fibers**

- Aerobic training increases recruitment of type 1 (slow-twitch) muscle fibers.
- Hypertrophy of type 1 muscle fibers, increasing their size by up to 25%.

- **Myoglobin Content and Capillary Supply**

- Increased myoglobin content enhances oxygen transport in the muscle.
- Enhanced capillary supply improves oxygen delivery to the muscle fibers.



Muscular Adaptations Due to Aerobic Training

- **Mitochondrial Adaptations**

- Increased number and size of mitochondria.
- Improved mitochondrial turnover, enhancing aerobic capacity.

- **Oxidative Enzymes**

- Increased mitochondrial oxidative enzyme activity.
- Enhanced fat oxidation enzymes improve fat utilization during submaximal exercise.
- Greater reliance on fat as a primary energy source during endurance activity.
- Sparing of muscle glycogen, preserving it for higher energy demands.



Musculoskeletal Adaptations Due to Anaerobic and Resistance Training

• Neuromuscular Adaptations

- Anaerobic training triggers significant adaptations in the nervous and muscular systems.
- **Central Nervous System Adaptations:**
 - Increased motor cortex activity at the cerebral cortex level, leading to better central adaptation.
 - Enhanced recruitment of fast-twitch motor units due to central adaptations.
- **Neuromuscular Junction:**
 - Adaptations at the neuromuscular junction improve communication between nerves and muscles.



Musculoskeletal Adaptations Due to Anaerobic and Resistance Training

● Timeline of Neural Adaptations

- Neural adaptations occur within the first 6 to 10 weeks of resistance or anaerobic training.
- After 10 weeks, muscle hypertrophy becomes more significant, contributing to strength gains.

● Muscle Hypertrophy

- Muscle hypertrophy refers to an increase in the cross-sectional area of the muscle.
- Growth of contractile proteins (actin and myosin) contributes to hypertrophy.
- Significant hypertrophy occurs after around 16 weeks of resistance training.



Musculoskeletal Adaptations Due to Anaerobic and Resistance Training

- **Muscle Fiber Type Adaptation**

- Anaerobic training primarily recruits type 2 (fast-twitch) muscle fibers, enhancing power and strength.

- **Connective Tissue Adaptations**

- Resistance training increases mechanical loading, stimulating bone formation (osteogenesis).
- Bone mineral density improves, leading to stronger bones with better mineral distribution.
- Bone adaptations generally take around 6 months of resistance training.

- **Bone-Specific Adaptations**

- Stress on specific muscles during training leads to targeted bone adaptation in corresponding bones.



The End

